

Osnovni pojmi

$$\mathbf{e}_\alpha = \frac{\partial \mathbf{r}}{\partial x_\alpha}$$
$$d\mathbf{r} = \sum \frac{\partial \mathbf{r}}{\partial x_\alpha} dx_\alpha$$
$$\mathbf{v} = \dot{\mathbf{r}} = \sum \dot{x}_\alpha \mathbf{e}_\alpha$$

Polarne koordinate

$$\mathbf{r} = r \cos \varphi \hat{\mathbf{i}} + r \sin \varphi \hat{\mathbf{j}}$$
$$\mathbf{e}_r = \cos \varphi \hat{\mathbf{i}} + \sin \varphi \hat{\mathbf{j}}$$
$$\mathbf{e}_\varphi = -r \sin \varphi \hat{\mathbf{i}} + r \cos \varphi \hat{\mathbf{j}}$$
$$|\mathbf{e}_r| = 1 \qquad |\mathbf{e}_\varphi| = r \qquad \mathbf{e}_r \cdot \mathbf{e}_\varphi = 0$$
$$\dot{\mathbf{e}}_r = \frac{\dot{\varphi}}{r} \mathbf{e}_\varphi \qquad \dot{\mathbf{e}}_\varphi = \frac{\dot{r}}{r} \mathbf{e}_\varphi - \dot{\varphi} r \mathbf{e}_r$$
$$T = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\varphi}^2)$$
$$\mathbf{a} = \mathbf{e}_r (\ddot{r} - \dot{\varphi}^2 r) + \mathbf{e}_\varphi (\ddot{\varphi} + 2 \frac{\dot{\varphi} \dot{r}}{r})$$

Sferične koordinate

$$\mathbf{r} = r \cos \varphi \sin \vartheta \hat{\mathbf{i}} + r \sin \varphi \sin \vartheta \hat{\mathbf{j}} + r \cos \vartheta \hat{\mathbf{k}}$$
$$\mathbf{e}_r = \cos \varphi \sin \vartheta \hat{\mathbf{i}} + \sin \varphi \sin \vartheta \hat{\mathbf{j}} + \cos \vartheta \hat{\mathbf{k}}$$
$$\mathbf{e}_\vartheta = r \cos \varphi \cos \vartheta \hat{\mathbf{i}} + r \sin \varphi \cos \vartheta \hat{\mathbf{j}} - r \sin \vartheta \hat{\mathbf{k}}$$
$$\mathbf{e}_\varphi = -r \sin \varphi \sin \vartheta \hat{\mathbf{i}} + r \cos \varphi \sin \vartheta \hat{\mathbf{j}}$$
$$|\mathbf{e}_r| = 1 \qquad |\mathbf{e}_\vartheta| = r \qquad |\mathbf{e}_\varphi| = r \sin \vartheta \qquad \mathbf{e}_\alpha \cdot \mathbf{e}_\beta = 0$$
$$T = \frac{1}{2} m (\dot{r}^2 + \dot{\varphi}^2 r^2 \sin^2 \vartheta + \dot{\vartheta}^2 r^2)$$

Neinercialni koordinatni sistemi

$$\frac{d\mathbf{e}'_\alpha}{dt} = \boldsymbol{\omega}' \times \mathbf{e}'_\alpha$$
$$\mathbf{r}' = \sum x'_\alpha \mathbf{e}'_\alpha = \mathbf{r} - \mathbf{R}$$
$$\mathbf{v}_{\text{rel}} = \sum \dot{x}'_\alpha \mathbf{e}'_\alpha$$
$$\dot{\mathbf{r}}' = \mathbf{v}_{\text{rel}} + \boldsymbol{\omega}' \times \mathbf{r}'$$
$$\ddot{\mathbf{r}}' = \mathbf{a}_{\text{rel}} + 2\boldsymbol{\omega}' \times \mathbf{v}_{\text{rel}} + \boldsymbol{\omega}' \times (\boldsymbol{\omega}' \times \mathbf{r}') + \dot{\boldsymbol{\omega}}' \times \mathbf{r}'$$
$$\mathbf{F}_s = -m\ddot{\mathbf{R}} - 2m\boldsymbol{\omega}' \times \mathbf{v}_{\text{rel}} - m\boldsymbol{\omega}' \times (\boldsymbol{\omega}' \times \mathbf{r}') - m\dot{\boldsymbol{\omega}}' \times \mathbf{r}'$$
$$m\mathbf{a}_{\text{rel}} = m \sum \ddot{x}_\alpha \mathbf{e}_\alpha = \sum \mathbf{F} + \mathbf{F}_s$$

Vrteči sistem (recimo na planetu)

$$\hat{\mathbf{i}}' = \cos(\Omega t) \hat{\mathbf{i}} + \sin(\Omega t) \hat{\mathbf{j}}$$
$$\hat{\mathbf{j}}' = -\sin(\Omega t) \sin \varphi \hat{\mathbf{i}} + \cos(\Omega t) \sin \varphi \hat{\mathbf{j}} + \cos \varphi \hat{\mathbf{k}}$$
$$\hat{\mathbf{k}}' = \sin(\Omega t) \cos \varphi \hat{\mathbf{i}} - \cos(\Omega t) \cos \varphi \hat{\mathbf{j}} + \sin \varphi \hat{\mathbf{k}}$$
$$\boldsymbol{\Omega} = \Omega \cos \varphi \hat{\mathbf{j}}' + \Omega \sin \varphi \hat{\mathbf{k}}'$$
$$\mathbf{R} = R \hat{\mathbf{k}}' \qquad \dot{\mathbf{R}} = \Omega \times \mathbf{R}$$
$$\ddot{\mathbf{r}} = \dot{\mathbf{v}}_{\text{rel}} + \boldsymbol{\Omega} \times (\dot{\mathbf{r}}' + \dot{\mathbf{R}}) = \mathbf{a}_{\text{rel}} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \boldsymbol{\Omega} \times [\boldsymbol{\Omega} \times (\mathbf{r}' + \mathbf{R})]$$
$$m\mathbf{a}_{\text{rel}} = \sum \mathbf{F} - 2m\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} - m\boldsymbol{\Omega} \times [\boldsymbol{\Omega} \times (\mathbf{r}' + \mathbf{R})]$$

Pomembne količine

1 delec

$$\mathbf{F} = m\ddot{\mathbf{r}}$$
$$\mathbf{F}_{21} = -\mathbf{F}_{12}$$

$$\mathbf{p} = m\mathbf{v}$$
$$\frac{d\mathbf{p}}{dt} = \mathbf{F}$$

$$\mathbf{l} = \mathbf{r} \times \mathbf{p} = m\mathbf{r} \times \mathbf{v}$$

$$\frac{d\mathbf{l}}{dt} = \mathbf{M} = \mathbf{r} \times \mathbf{F}$$

$$A = \int_\gamma \mathbf{F} \cdot d\mathbf{r}$$

$$T = \frac{1}{2} m \mathbf{v} \cdot \mathbf{v}$$

Sila je konzervativna, če $\mathbf{F} = -\boldsymbol{\nabla} U$.

$\mathbf{F}$ konzervativna $\iff$ $\frac{\partial F_x}{\partial y} = \frac{\partial F_y}{\partial x}$ (v 3D po parih xy , yz , xz).

$\mathbf{F}$ konzervativna $\iff$ $A = \int_\gamma \mathbf{F} \cdot \mathbf{r} = -\Delta U$ neodvisno od poti.

Sistem N delcev

$$\mathbf{r}_T = \sum \frac{m_i \mathbf{r}_i}{\sum m_i}$$
$$\mathbf{P} = \sum \mathbf{p}_i = M \dot{\mathbf{r}}_T$$
$$\frac{d\mathbf{P}}{dt} = M \mathbf{a}_T = \mathbf{F}^{(z)}$$
$$\mathbf{L} = \sum \mathbf{l}_i = \mathbf{r}_T \times M \mathbf{v}_T + \sum \mathbf{r}'_i \times m \mathbf{v}'_i$$
$$\dot{\mathbf{L}} = \mathbf{M}^{(z)} + \mathbf{M}^{(n)} = \sum \mathbf{r}_i \times \mathbf{F}_i^{(z)} + \sum_{j < i} (\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{F}_{ij}$$

Če so sile centralne $\left(\mathbf{F}_{ij} = F_{ij} \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|}\right)$, je $\mathbf{M}^{(n)} = 0$.

$$A = A^{(z)} + A^{(n)}$$
$$T = \frac{1}{2} M v_T^2 + \sum \frac{1}{2} m_i v_i'^2$$
$$V = \sum_i V_i + \frac{1}{2} \sum_{ij, i \neq j} V_{ij}$$

Virialni teorem (sistem N delcev)

$$2\langle T \rangle = \frac{1}{2} \langle \sum_{ij, i \neq j} \frac{\partial V_{ij}}{\partial r_{ij}} r_{ij} \rangle - \langle \sum_i \mathbf{F}_i^z \cdot \mathbf{r}_i \rangle$$
$$V(\mathbf{r}) = \alpha r^n \implies 2\langle T \rangle = n \langle V \rangle$$

Lagrangev formalizem

$$\sum_{i=1}^N \left(\mathbf{F}_i^{(a)} - m_i \mathbf{a}_i\right) \delta \mathbf{r}_i = 0 \qquad \text{(D'Alembertov princip)}$$

$$Q_j = \sum_{i=1}^N \mathbf{F}_i^{(a)} \frac{\partial \mathbf{r}_i}{\partial \dot{q}_j}$$
$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_j} - \frac{\partial T}{\partial q_j} = Q_j$$

Če $\mathbf{F}_i = -\boldsymbol{\nabla} V$, $\forall i$:

$$L = T - V$$
$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_j} - \frac{\partial L}{\partial q_j} = 0$$

Če $\frac{\partial L}{\partial q_i} = 0$, potem $\frac{\partial L}{\partial \dot{q}_i} = konst. = p_i$ (posplošeni impulz, q_i pravimo ciklična koordinata).

Če $\frac{\partial L}{\partial t} = 0$, potem $H = \sum_j \dot{q}_j \frac{\partial L}{\partial \dot{q}_j} - L = konst..$

Če $V \neq V(\dot{q}_1, \dots, \dot{q}_n, t)$, potem $H = T + V$.

Lagrangeva funkcija je nedoločena do $\frac{d}{dt} F(q_1, \dots, q_n, t)$.

Izrek Emmy Noether

Imejmo enoparametrično preslikavo koordinat $q_i \mapsto Q_i(s, t)$.

Ta ustreza zvezni simetriji L , če $\frac{\partial L(Q_i(s, t), \dot{Q}_i(s, t), t)}{\partial s} = 0$.

Tedaj obstaja ohranjena količina $\sum_i \frac{\partial L}{\partial \dot{q}_i} \frac{\partial \dot{Q}_i}{\partial s} |_{s=0}$.

Hamiltonov princip

$$S(L) = \int_{t_1}^{t_2} L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t) dt \qquad \text{(akcija)}$$

Enačbe gibanja ustrezajo minimumu funkcionala akcije S .

Problem dveh teles

Imamo $m_1, m_2, \mathbf{r}_1, \mathbf{r}_2$.

$$M \mathbf{r}_t = (m_1 + m_2) \mathbf{r}_t = m_1 \mathbf{r}_1 + m_2 \mathbf{r}_2$$

$$m = \frac{m_1 m_2}{m_1 + m_2}$$

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1 \implies \mathbf{r}_{1,2} = \mathbf{r}_t \mp \frac{m_{2,1}}{M} \mathbf{r}$$

$$L = \frac{1}{2} M v_t^2 + \frac{1}{2} m |\dot{\mathbf{r}}|^2 - V(\mathbf{r}, \dot{\mathbf{r}})$$
$$\mathbf{p}_t = M \mathbf{v}_t = konst.$$

Centralni potencial

$$\text{Imamo } V = V(r).$$
$$\mathbf{F} = -\boldsymbol{\nabla} V = -\frac{dV}{dr} \frac{\mathbf{r}}{r}$$
$$\mathbf{L} = konst.$$
$$\mathbf{L} \cdot \mathbf{r} = 0 \implies \text{Gibanje je ravninsko.}$$

$$\text{Definiramo } \mathbf{L} = L \hat{e}_z.$$
$$L = \frac{1}{2} m (\dot{r}^2 + r^2 \dot{\varphi}^2) - V(r)$$
$$p_\varphi = m r^2 \dot{\varphi} = l = konst.$$
$$\frac{dS}{dt} = \frac{1}{2} r^2 \dot{\varphi} = konst. \qquad \text{(2. Keplerjev zakon)}$$

$$V_{\text{ef}} = \frac{p_\varphi^2}{2mr^2} + V(r)$$
$$H = \frac{1}{2} m \dot{r}^2 + V_{\text{ef}}$$
$$\frac{1}{2} m \dot{r}^2 \geq 0 \implies H \geq V_{\text{ef}} \implies \text{kvalitativna določitev orbit.}$$

Pogosto uporabna substitucija $u = \frac{1}{r}$.

Pogosto uporabno $\frac{d}{dt} = \frac{d\varphi}{dt} \frac{d}{d\varphi} = \frac{p_\varphi}{mr^2} \frac{d}{d\varphi}$

Keplerjev problem

$$\text{Imamo } V(r) = -\frac{k}{r}.$$
$$\mathbf{A} = \mathbf{p} \times \mathbf{L} - m k \frac{\mathbf{r}}{r} = konst. \qquad \text{(Laplace-Runge-Lenzov vektor)}$$
$$\mathbf{A} \cdot \mathbf{L} = 0$$

$$r(\varphi) = \frac{p_\varphi^2 / \mu k}{1 - \sqrt{1 + 2 p_\varphi^2 H / \mu k^2} \cos(\varphi - \varphi_0)} = \frac{r_0}{1 - \varepsilon \cos(\varphi - \varphi_0)}$$

$$\varepsilon = \sqrt{1 + \frac{2 p_\varphi^2 H}{k^2 \mu}}$$

$$\varepsilon = 0 \rightarrow H = -\frac{mk^2}{2I^2} \text{ - krožnica,} \qquad \varepsilon < 1 \rightarrow H < 0 \text{ - elipsa}$$
$$\varepsilon = 1 \rightarrow H = 0 \text{ - parabola,} \qquad \varepsilon > 1 \rightarrow H > 0 \text{ - hiperbola}$$

Orbite za $V(r) = \frac{k}{r^2}$ -npr. nabiti delci

$$V_{eff} = \frac{p_\varphi^2}{2\mu} (1 - \gamma) \frac{1}{r^2}$$

$$\mathbf{1.} \quad \gamma < 1: \alpha^2 = 1 - \gamma$$
$$r(\varphi) = \frac{1}{A \cos(\alpha(\varphi - \varphi_0))}, \quad A = r_{min}, \quad \Delta\varphi = \alpha\pi$$

$$\mathbf{2.} \quad \gamma = 1: \text{kroženje}$$
$$r(\varphi) = \frac{1}{A \cos(\varphi - \varphi_0) + B}$$

$$\mathbf{3.} \quad \gamma > 1: -\alpha^2 = 1 - \gamma$$
$$r(\varphi) = \frac{1}{A e^{-\alpha\varphi} + B e^{\alpha\varphi}}$$

Orbite za $V(r) = k r^2$ -npr. vzmet

$$x(t) = x_0 \cos(\omega t + \delta_x),$$

$$y(t) = y_0 \sin(\omega t + \delta_y),$$

Sipanje delcev

$$dI_{\text{vh}} = j dS = j 2\pi b db$$

$$dI_{\text{iz}} = j \sigma(\Omega) d\Omega$$

$$\sigma(\vartheta) = \left. \frac{b}{\sin \vartheta} \right| \frac{db}{d\vartheta} \Big| \qquad \text{(diferencialni sipalni presek)}$$

$$\sigma_{\text{tot}} = \int \sigma(\Omega) d\Omega = \pi b_{\text{max}}^2$$

Togo telo

$|\mathbf{r}_i - \mathbf{r}_j| = r_{ij} = konst.$

Eulerjevi koti

$$T(\varphi) = \begin{bmatrix} \cos \varphi & \sin \varphi & 0 \\ -\sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{precesija, } \mathbf{r} \rightarrow \mathbf{r}'')$$

$$U(\vartheta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \vartheta & \sin \vartheta \\ 0 & -\sin \vartheta & \cos \vartheta \end{bmatrix} \quad (\text{nutacija, } \mathbf{r}'' \rightarrow \mathbf{r}''')$$

$$V(\psi) = \begin{bmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{rotacija, } \mathbf{r}''' \rightarrow \mathbf{r}')$$

$$R = V(\psi)U(\vartheta)T(\varphi) \quad (\text{pasivna rotacija})$$

Za vsako rotacijo $R \exists \mathbf{n} \neq 0 : R\mathbf{n} = \mathbf{n}$, torej obstaja os rotacije.

V lastnem sistemu S' :

$$\boldsymbol{\omega} = \dot{\varphi} \mathbf{e}_3 + \dot{\vartheta} \mathbf{e}'_1 + \dot{\psi} \mathbf{e}'_3 = (\sin \vartheta \sin \psi \dot{\varphi} + \cos \psi \dot{\vartheta}) \mathbf{e}'_1 +$$

$$(\sin \vartheta \cos \psi \dot{\varphi} - \sin \psi \dot{\vartheta}) \mathbf{e}'_2 + (\cos \vartheta \dot{\varphi} + \dot{\psi}) \mathbf{e}'_3$$

Togo telo z 1 nepremično točko

Po Eulerjevem izreku je gibanje takega telesa ob vsakem času rotacija okoli neke osi.

$$\frac{d\mathbf{r}}{dt} = \boldsymbol{\omega} \times \mathbf{r}$$

$$\mathbf{L} = \sum_i m_i [(\mathbf{r}_i \cdot \mathbf{r}_i) \boldsymbol{\omega} - (\mathbf{r}_i \cdot \boldsymbol{\omega}) \mathbf{r}_i] = \underline{J} \boldsymbol{\omega}$$

$$\underline{J} = \int \rho dV \begin{bmatrix} y^2 + z^2 & -xy & -xz \\ -yx & x^2 + z^2 & -yz \\ -zx & -zy & x^2 + y^2 \end{bmatrix}$$

$$\mathbf{L}' = R\mathbf{L} \text{ in } \boldsymbol{\omega}' = R\boldsymbol{\omega} \implies \underline{J}' = R\underline{J}R^T$$

$$T = \frac{1}{2} \sum_i m_i v_i^2 = \frac{1}{2} \boldsymbol{\omega}' \underline{J} \boldsymbol{\omega}'$$

Delajmo v lastnem sistemu telesa.

$$\frac{\delta \mathbf{L}}{\delta t} = \left(\frac{d\mathbf{L}}{dt} \right)_{\text{rel}} + \boldsymbol{\omega} \times \mathbf{L} = \mathbf{M}$$

$$\left(\frac{d\mathbf{L}}{dt} \right)_{\text{rel}} = \sum_{\alpha} J_{\alpha\alpha} \dot{\omega}_{\alpha} \mathbf{e}_{\alpha}$$

Eulerjeve enačbe

$$J_1 \dot{\omega}_1 - (J_2 - J_3) \omega_2 \omega_3 = M_1$$

$$J_2 \dot{\omega}_2 - (J_3 - J_1) \omega_3 \omega_1 = M_2$$

$$J_3 \dot{\omega}_3 - (J_1 - J_2) \omega_1 \omega_2 = M_3$$

Vpeta osno simetrična vrtavka

Uporabljamo Eulerjeve kote.

$$J_1 = J_2 \neq J_3$$

$$L = \frac{1}{2} J_1 (\omega_1^2 + \omega_2^2) + \frac{1}{2} J_3 \omega_3^2 - mgz = \\ = \frac{1}{2} J_1 (\sin^2 \vartheta \dot{\varphi}^2 + \dot{\vartheta}^2) + \frac{1}{2} J_3 (\dot{\psi} + \cos \vartheta \dot{\varphi})^2 - mgl \cos \vartheta$$

$$p_{\psi} = J_3 (\dot{\psi} + \cos \vartheta \dot{\varphi}) = J_3 \omega_3 = konst = J_1 a$$

$$p_{\varphi} = (J_1 \sin^2 \vartheta + J_3 \cos^2 \vartheta) \dot{\varphi} + J_3 \cos \vartheta \dot{\psi} = konst. = J_1 b$$

$$E = T + V = konst.$$

$$\dot{\varphi} = \frac{b-a \cos \vartheta}{\sin^2 \vartheta}$$

$$\dot{\psi} = \frac{J_1}{J_3} a - \frac{b-a \cos \vartheta}{\sin^2 \vartheta} \cos \vartheta$$

Osnovna enačba vrtavke

$$\tilde{E} = \frac{1}{2} J_1 \dot{\vartheta}^2 + \tilde{V}(\vartheta) = E - \frac{1}{2} J_3 \omega_3^2 = konst.$$

$$\tilde{V}(\vartheta) = \frac{1}{2} \frac{(b-a \cos \vartheta)^2}{\sin^2 \vartheta} + mgl \cos \vartheta$$

$$\implies \dot{\vartheta}^2 = \frac{2}{J_1} (\tilde{E} - \tilde{V}(\vartheta))$$

Pogosto uporabna substitucija $u = \cos \vartheta$.

$$\alpha = \frac{2\tilde{E}}{J_1} \quad \text{in} \quad \beta = \frac{2mgl}{J_1}$$

$$t = \sqrt{\frac{J_1}{2}} \int_{\vartheta(0)}^{\vartheta(t)} \frac{d\vartheta}{\sqrt{\tilde{E} - \tilde{V}(\vartheta)}} = - \int_{u(0)}^{u(t)} \frac{du}{\sqrt{f(u)}}$$

$$f(u) = (1 - u^2)(\alpha - \beta u) - (b - au)^2$$

S pogojem $f(u) \geq 0$ kvalitativno določimo obnašanje vrtavke.

Hamiltonov formalizem

Namesto $s \dot{q}_i$ delamo s $p_i = \frac{\partial L}{\partial \dot{q}_i}$.

$$\dot{p}_i = \frac{\partial L}{\partial q_i} \quad (\text{iz EL enačbe})$$

Hamiltonove enačbe

$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}$$

Nabit delec v magnetnem polju

V Lagrangevemu formalizmu lahko upoštevamo tudi potenciale

$$U \text{ za katere velja } Q_i = -\frac{\partial U(q_i, \dot{q}_i, t)}{\partial q_i} + \frac{d}{dt} \frac{\partial U(q_i, \dot{q}_i, t)}{\partial \dot{q}_i}.$$

$$\mathbf{B} = \nabla \times \mathbf{A}, \quad \mathbf{E} = -\nabla \varphi - \frac{\partial \mathbf{A}}{\partial t}$$

$$U = e\varphi - e\mathbf{v} \cdot \mathbf{A}$$

$$\mathbf{F} = e(\mathbf{E} + \mathbf{v} \times \mathbf{B}) = -\nabla_{\mathbf{r}} U + \frac{d}{dt} \nabla_{\mathbf{v}} U$$

$$L = \frac{1}{2} m v^2 - U = \frac{1}{2} m v^2 - e\varphi + e\mathbf{v} \cdot \mathbf{A}$$

$$\mathbf{p} = \nabla_{\mathbf{v}} L = m\mathbf{v} + e\mathbf{A}$$

$$H = \mathbf{p} \cdot \mathbf{v} - L = \frac{1}{2} m v^2 + e\varphi = \frac{|\mathbf{p} - e\mathbf{A}|^2}{2m} + e\varphi$$

Poissonovi oklepaji

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right)$$

$$\{f, \lambda g + \mu h\} = \lambda \{f, g\} + \mu \{f, h\}, \quad \lambda, \mu \in \mathbb{C}$$

$$\{f, g\} = -\{g, f\}$$

$$\{f, gh\} = \{f, g\}h + \{f, h\}g$$

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0$$

$$\frac{df}{dt} = \{f, H\} + \frac{\partial f}{\partial t}$$

$$f \text{ konstanta gibanja } \left(\frac{df}{dt} = 0 \right) \implies \{H, f\} = \frac{\partial f}{\partial t}$$

$$\{q_i, q_j\} = 0$$

$$\{p_i, p_j\} = 0$$

$$\{q_i, p_j\} = \delta_{ij}$$

$$\{l_i, l_j\} = \varepsilon_{ijk} l_k$$

Majhna nihanja

$$L = \frac{1}{2} \sum_{ij} w_{ij}(\underline{q}) \dot{q}_i \dot{q}_j - V(\underline{q})$$

Naj bo $\underline{q}^0$ ravnovesne lega sistema in $\underline{\eta} = \underline{q} - \underline{q}^0$.

$$\implies V(\underline{q}) = V(\underline{q}^0) + \frac{1}{2} \sum_{ij} \frac{\partial^2 V}{\partial q_i \partial q_j} \Big|_{\underline{q}^0} \eta_i \eta_j + \dots$$

$$\implies w_{ij}(\underline{q}) = w_{ij}(\underline{q}^0) + \dots$$

$$\text{Definiramo } T_{ij} = w_{ij}(q_0) \text{ in } V_{ij} = \frac{\partial^2 V}{\partial q_i \partial q_j} \Big|_{\underline{q}^0}.$$

$$\implies \tilde{L} = T - (V - V_0) = \frac{1}{2} (\dot{\underline{\eta}}^T \underline{T} \dot{\underline{\eta}} - \underline{\eta}^T \underline{V} \underline{\eta})$$

$$\implies \sum_j T_{ij} \ddot{\eta}_j + \sum_j V_{ij} \eta_j = 0$$

Lastna nihanja $\eta_i = C \alpha_i e^{i\omega t}$

$$\implies \underline{V} \underline{a} = \omega^2 \underline{T} \underline{a} \implies \det \left(\underline{V} - \omega^2 \underline{T} \right) = 0 \implies$$

$$\implies \text{Lastne frekvence } \omega_k \text{ in lastni vektorji } \underline{a}_k \text{ } (\underline{A}).$$

Splošno nihanje $\eta_i(t) = \sum_{k=1}^n a_{ki} \alpha_k(t)$

Izberemo $\underline{a}_k$ tako, da $\underline{a}_i^T \underline{T} \underline{a}_j = \delta_{ij}$

$$\implies \underline{A}^T \underline{T} \underline{A} = \underline{I} \quad (\text{posplošena ortogonalnost}) \text{ in}$$

$$\underline{A}^T \underline{V} \underline{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$$

$$\implies \alpha_k = \begin{cases} C_k \cos(\omega_k t + \delta_k); & \omega_k \neq 0 \\ C_k + \tilde{C}_k t; & \omega_k = 0 \end{cases}$$

α_i so normalne koordinate

$$L = \frac{1}{2} \sum_i (\dot{\alpha}_i^2 - \omega_i^2 \alpha_i^2)$$

$$E = \frac{1}{2} \sum_i (\dot{\alpha}_i^2 + \omega_i^2 \alpha_i^2) = \sum_i E_i$$

Vztrajnostni momenti

Valj

$$J_x = J_y = \frac{1}{4} m r^2 + \frac{1}{12} m h^2, \quad J_z = \frac{1}{2} m r^2$$

Plašč valja

$$J_x = J_y = \frac{1}{2} m r^2 + \frac{1}{12} m h^2, \quad J_z = m r^2$$

Tanka palica

$$J_x = J_y = \frac{1}{12} m l^2$$

Stožec

$$J_x = J_y = \frac{3}{20} m (r^2 + \frac{1}{4} h^2), \quad J_z = \frac{3}{10} m r^2$$

Kvadratna piramida

$$J_x = J_y = \frac{1}{20} m (a^2 + \frac{3}{4} h^2), \quad J_z = \frac{1}{10} m a^2$$

Krogla

$$J_x = J_y = J_z = \frac{2}{5} m r^2$$

Krogelna lupina

$$J_x = J_y = J_z = \frac{2}{3} m r^2$$

Votla krogla

$$J_x = J_y = J_z = \frac{2}{5} m \frac{R^5 - r^5}{R^3 - r^3}$$

Torus

$$J_x = J_y = \frac{1}{8} m (4R^2 + 5r^2), \quad J_z = m (R^2 + \frac{3}{4} r^2)$$

Kvader

$$J_x = \frac{1}{12} m (b^2 + c^2), \quad J_y = \frac{1}{12} m (a^2 + c^2), \quad J_z = \frac{1}{12} m (a^2 + b^2)$$